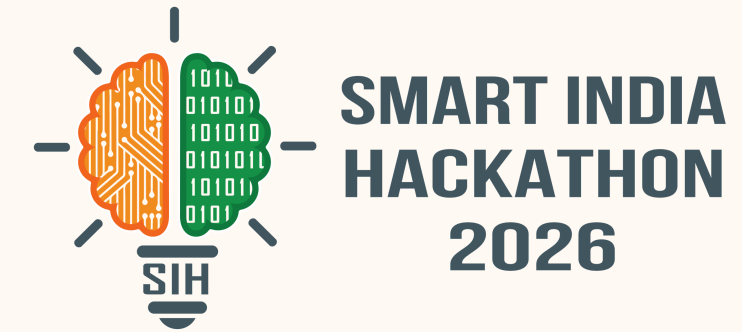


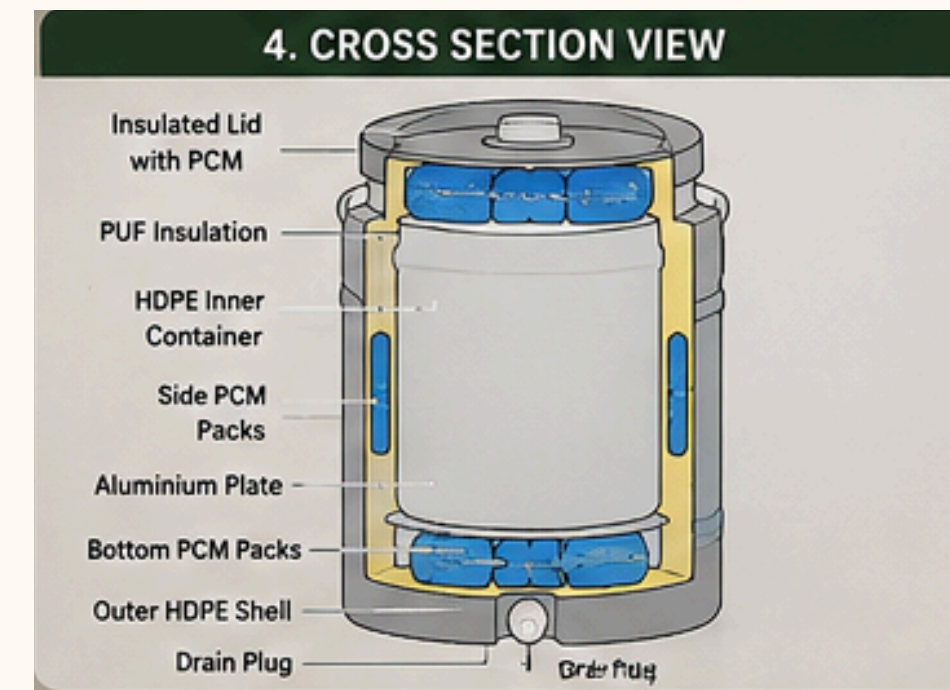
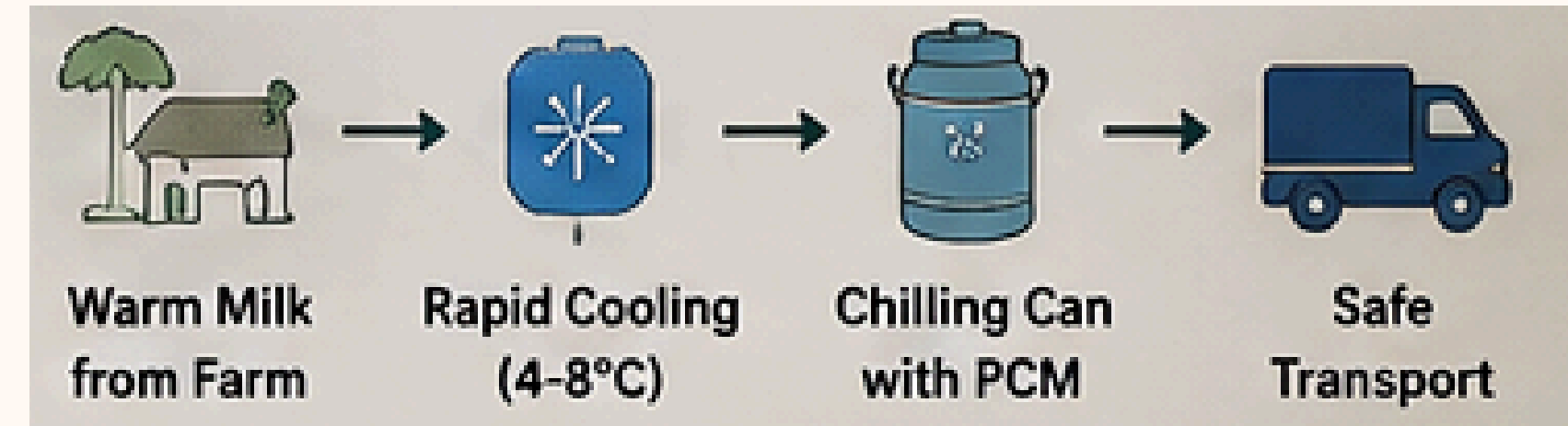


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- Problem Statement ID – 26110
- Problem Statement Title- Development of a Low cost Light weight Milk Chilling Can for Small Scale Dairy Farmers
- Theme- Agriculture, FoodTech and Rural Development
- PS Category- Hardware
- Team ID- SIH26-A0H-T192
- Team Name (Registered on portal) Kilowatt Krew





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PROBLEMS IN HAND

MILK QUALITY:Temperature rise can accelerate microbial growth and deterioration of milk quality.Inadequate temperature control between collection and chilling can reduce the quality of milk reaching the processing facility.

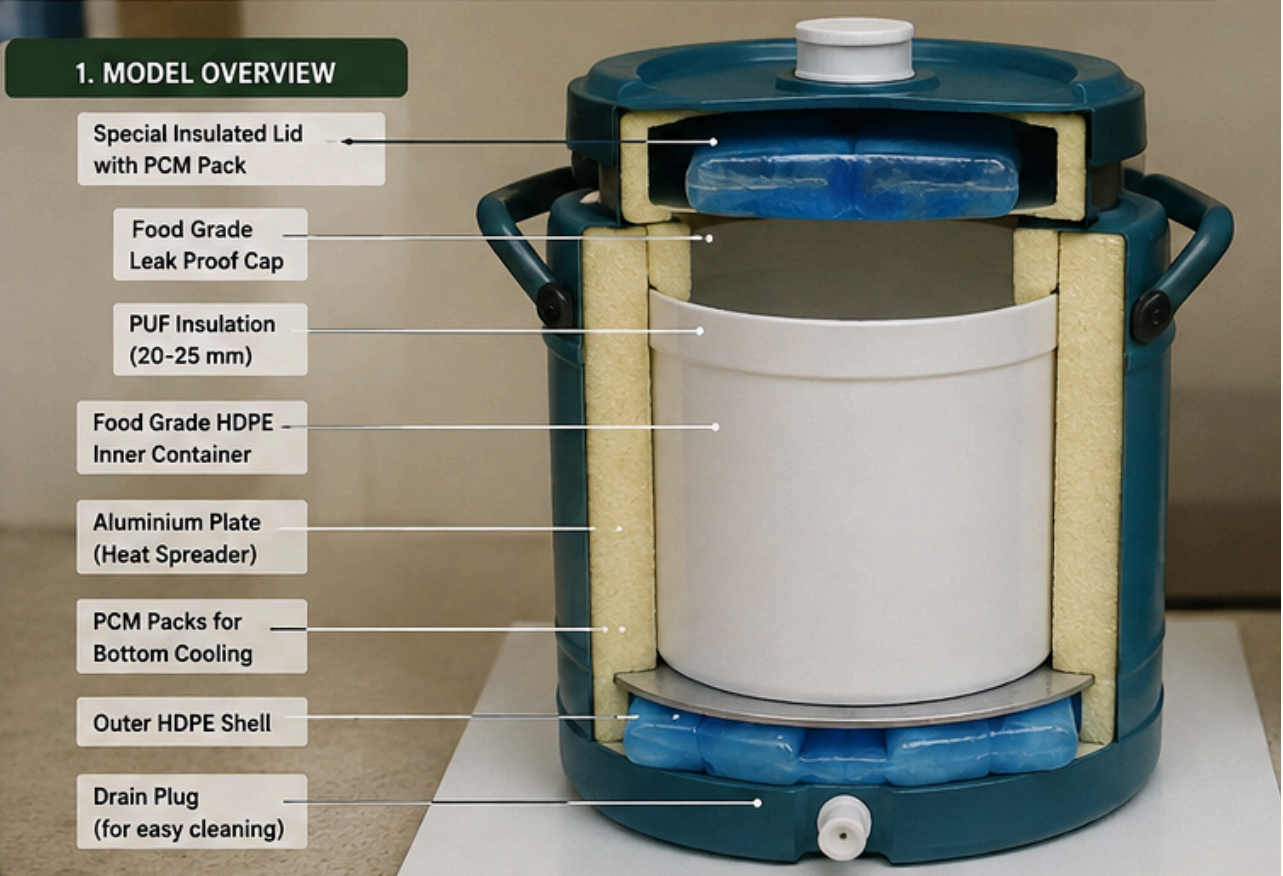
Mechanical:Rough roads and hilly terrain expose milk cans to vibration and shocks.The current existing can cant withstand the pressure and which build heat inside the can

Thermal: Maintaining milk at the required 4–8°C during transportation is challenging.Milk is exposed to ambient heat during collection and transit.Long transportation durations increase the cooling requirement.

Refrigerant Impact: Conventional bulk chillers rely extensively on chemical refrigerants like Freon or Ammonia, which carry high global warming potential and fail to offer an eco-friendly option for smaller operations.

PROPOSED SOLUTION DAIRY CHILL

LOW-COST LIGHTWEIGHT MILK CHILLING CAN For Safe Milk Transportation in Rural Areas



Why We Stand Out

We don't just transport milk – we actively protect its quality throughout transportation.”

- **Active + Passive Cooling** – cooling system brings the temperature down, while insulation and PCM help maintain it.
- **Temperature Monitoring** – enables continuous awareness of milk temperature.
- **Transport Protection** – shock-absorbing construction improves durability on rough roads.
- **Designed for Rural & Hilly Transport** – focuses on portability, ruggedness and practical field conditions.



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KEY FEATURES

- **Active Cooling System** – rapidly brings milk toward the required 4–8°C range.
- **PCM-Based Thermal Retention** – helps maintain temperature and reduces dependence on continuous active cooling.
- **Real-Time Temperature Monitoring** – continuously monitors the milk temperature.
- **Shock-Resistant Construction** – designed to withstand vibration and rough transportation conditions.
- **Energy-Conscious Operation** – insulation + PCM reduce the cooling load on the active system.
- **Multi-Layer Thermal Insulation** – minimizes heat gain from the surroundings.



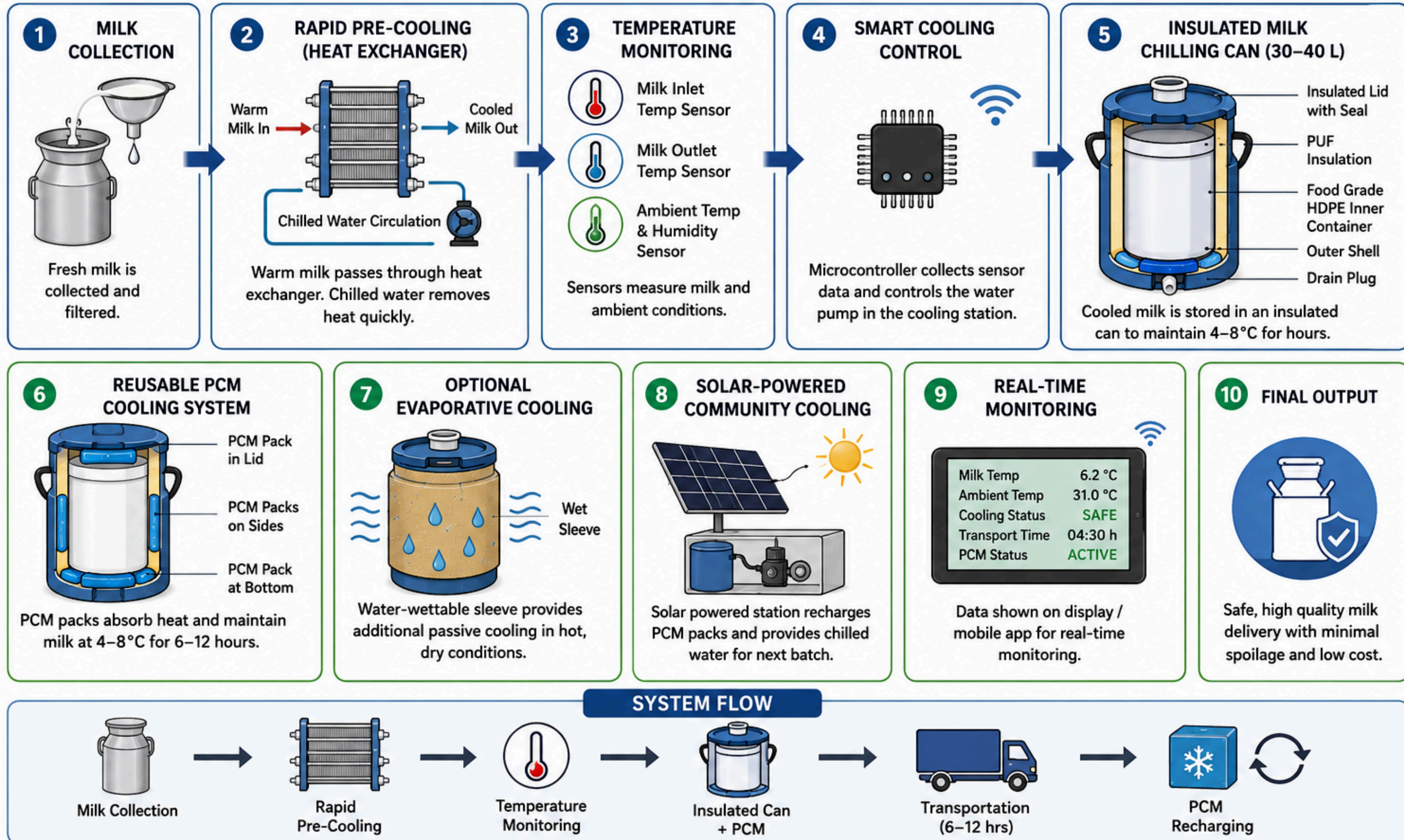
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TECHNICAL APPROACH

Low-Cost Passive Milk Chilling & Transportation System



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MILK CHILLING CAN – COST COMPARISON & ACCESSIBILITY

Affordable, Passive Cooling Solution for Rural Dairy Supply Chain

1. COST COMPARISON – 30–40 L CAPACITY

FEATURE / MATERIAL	CONVENTIONAL ALUMINIUM CAN	CONVENTIONAL STAINLESS STEEL (SS) CAN	OUR PROPOSED MILK CHILLING CAN
 Capacity	30–40 L	30–40 L	30–40 L
 Inner Container Material	Aluminium	Stainless Steel	Food-grade HDPE
 Insulation	×	×	✓ PUF (20–25 mm)
 Cooling System	×	×	✓ Reusable PCM / Ice Packs
 Special Cooling Lid	×	×	✓
 Side / Bottom Cooling	×	×	✓
 Approx. Can Price	₹2,400 – ₹4,500	₹4,650 – ₹10,500+	₹3,300 (Target Cost)
 Electricity During Transport	No	No	No (Passive Cooling)
 Temperature Maintenance	Poor	Poor	Designed for 4–8°C*
 Weight	Medium	High	Lower (Lightweight Design)
 Reusable Cooling Packs	×	×	✓ Yes
 Cleaning	Easy	Easy	Easy (Removable Modules)
 Rural Transport Suitability	Good	Good	Very Good

OUR PROPOSED CAN – COST BREAKDOWN (Target: ₹3,300)

COMPONENT / MATERIAL	ESTIMATED COST (₹)
Food-grade HDPE inner container	850
Outer HDPE / Protective shell	400
PUF Insulation (20–25 mm)	450
PCM / Gel Cooling Packs (Reusable)	700
Aluminium Heat Spreader Plates	250
Special Insulated Lid + Sealing	300
Handles, Drain Plug, Fittings	200
Temperature Indicator	100
Miscellaneous / Assembly	50
TOTAL TARGET COST	₹ 3,300



2. HOW FARMERS CAN ACCESS THIS CAN

We make it affordable and accessible!



A. DAIRY COOPERATIVE SUBSIDIES / FINANCING

- Milk cooperative societies purchase the cans in bulk.
- Distributed to member farmers.
- Cost recovered through micro-deductions from weekly milk payments.
- Low monthly deduction → Easy for farmers to afford.

Example:	Cost of Can (Target)	: ₹3,300
	Cooperative Subsidy (Avg.) (25–40%)	: ₹825 – ₹1,320
	Farmer Pays Out of Pocket	: ₹1,980 – ₹2,475
	Recovery Period (via deductions)	: 6 – 10 months



B. GOVERNMENT SCHEMES – POSSIBLE SUPPORT

Our project fits well with Government schemes focused on dairy development, rural livelihoods & post-harvest management.

Scheme Name	Implementing Ministry / Agency	What It Offers	How It Can Help Our Project
National Programme for Dairy Development (NPDD)	Department of Animal Husbandry & Dairying (DAHD), Govt. of India	Subsidy for dairy infrastructure, milk chilling equipment, capacity building	Our cans can be included under subsidy for milk chilling & transport solutions
Dairy Entrepreneurship Development Scheme (DEDS)	NABARD	Subsidy / financial assistance up to 25% of project cost	Cooperatives / SHGs can apply to purchase cans in bulk
Rashtriya Gokul Mission (RGM)	DAHD, Govt. of India	Support for infrastructure and value chain development	Strengthens milk value chain with better cooling & transport solutions
Mission for Integrated Development of Horticulture (MIDH) – Post Harvest Management	Ministry of Agriculture & Farmers Welfare	Support for post-harvest technologies & value addition	Can be considered under post-harvest management for perishable produce like milk (in some states)
North East Region Dairy Development Scheme (NERDDS)	DAHD (Special focus on NE States)	Infrastructure & value chain support in NE states	Highly relevant for our target region (Northeast India)

KEY BENEFITS OF SCHEMES



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IMPACT AND BENEFITS



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HIGHER FARMER INCOME

- Less milk rejection due to poor quality
- Better milk quality can improve the value received by small and marginal dairy farmers
- Aims to increase farmers revenue by 5 to 15 %
- Stable 4 to 8 C Cooling preserves milk fat lipids which are highly sensitive to temperature changes

LIGHTWEIGHT & AFFORDABLE

- Lightweight HDPE/ composite construction reduce handling and transportation effort
- Designed to be significantly more economical than conventional insulated chilling container
- Our target full 40 Lit can approximately weighs 49 -53 kg which is approximately the same weight as the usual milk cans
- Our hybrid can's target cost is around 3300 Rs



3. ENERGY FREE OPERATION

- No electricity required during transport.
- Works on passive cooling using ice/PCM and insulation.



1. IMPROVED MILK QUALITY

Maintains milk at safe temperature (4–8°C), reducing bacterial growth and spoilage. Results in better taste, higher shelf life and premium pricing.



2. REDUCED MILK SPOILAGE AND LOSSES

Minimizes quality degradation during collection and transportation. Reduces quantity and financial losses for farmers.

BETTER RURAL MILK SUPPLY CHAIN

- Enables safe transportation from Farm to village collection centre to dairy facility
- Reduces dependence on expensive bulk milk cooling infrastructure at every farm

SUSTAINABLES & REUSEABLE

- Reusable cooling cartridges reduce dependence on disposable ice
- A shared solar-power/community cooling station can recharge the cooling modules for repeated use

ELECTRICITY -INDEPENDENT TRANSPORTATION

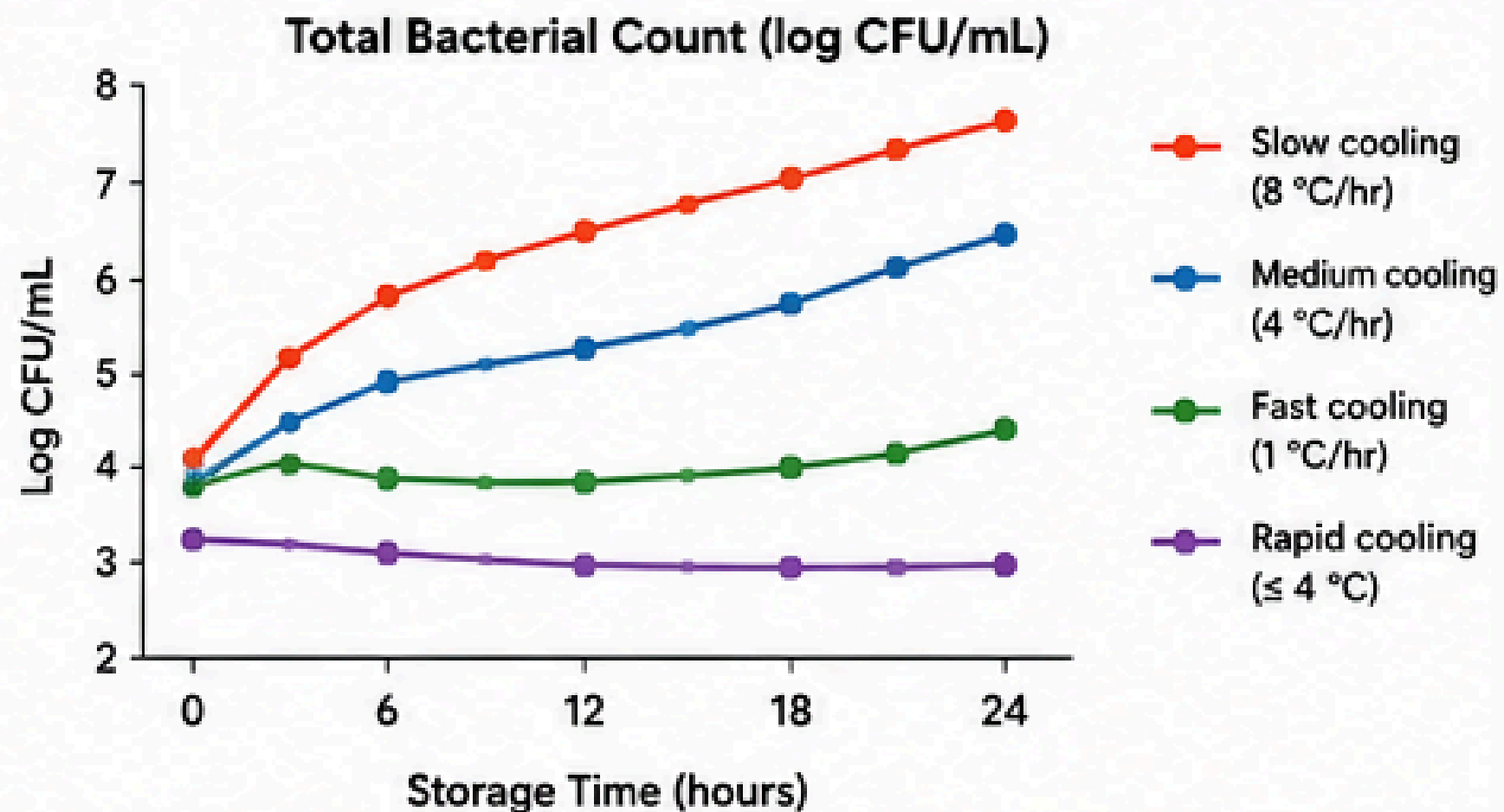
- Reusable PCM/ice cartridges maintain the temperature without continuous power during transportation
- Particularly useful in remote hilly and North-Eastern regions with unreliable electricity



4. EASY TO CLEAN AND MAINTAIN

- Smooth food-grade inner surface.
- Leak-proof design ensures hygiene and easy washing.

Paludetti, L. F. et al. (2018). The effect of different precooling rates and cold storage on milk microbiological quality and composition. *Journal of Dairy Science*.



3.Ndyabawe, K. & Kisaalita, W. S. (2014). Diffusion of an evaporative cooler innovation among smallholder dairy farmers of Western Uganda. *Technology in Society*.

Chaomuang, N. et al. (2025). Thermal performance evaluation of water-based gel packs and insulating covers for raw milk preservation. *Applied Food Research*

Thermal Performance Results

Cooling Setup	Initial Milk Temp (°C)	Milk Temp After 8 h (°C)	Temp Range Maintained
No Cooling (Control)	30.2	18.6	Poor
Ice Only	30.1	6.2	Moderate
Gel Pack Only	30.0	5.4	Good
Gel Pack + Insulating Cover	30.1	4.3	Excellent (4–8 °C)

<https://www.sciencedirect.com/science/article/pii/S2772502225005621?via%3Dihub>

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<https://epubs.icar.org.in/index.php/AAZ/article/view/134488>